

Seungyeop Lee

Software Engineer, Samsung Electronics

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Research Interests

Lower-limb wearable robots, vision-based perception, and human-robot interaction

Education

M.S., Mechanical Engineering; Integrated Major in Intelligent Mobility

Korea University | Seoul, Republic of Korea | Sep. 2022 - Feb. 2025 | GPA: 4.45/4.5

- Research Lab: Human-Machine Systems Lab (Prof. Shinsuk Park).
- Thesis: A Development of Intuitive Controller and Monocular Vision based Interface for Disaster Response UAV.

B.S., Mechanical Engineering

Korea University | Seoul, Republic of Korea | Mar. 2016 - Aug. 2022 | GPA: 3.45/4.5

- Absence from Oct. 2017 to Jun. 2019 due to military service.

Research Experience

Software Engineer, Spatial AI Development

Robot Vacuum Cleaner SW Lab - Perception Team, Samsung Electronics | Suwon, South Korea | May 2026 - Present

Spatial AI for Resource-Constrained Robot Vacuum Cleaners

- Contributed to a dual-process robot intelligence architecture combining task knowledge reuse and incremental scene-graph enrichment to reduce repeated reasoning on resource-constrained robots.
- Developed a viewpoint selection algorithm that selects navigation coordinates for robots to acquire object information based on visibility and accessibility.
- Developed room classification using object occurrence likelihoods and spatially accumulated visual evidence from observations collected during navigation.
- Integrated VLM, perception, and navigation into a unified Spatial AI system and validated in NVIDIA Isaac Sim.

Research Intern

Seoul National University, Wearable Robotics Laboratory | Seoul, South Korea | Advisor: Prof. Jinsoo Kim | Dec. 2024 - Feb. 2025

Embedded Controller and Sensor Integration for a Soft Exosuit

- Developed the Jetson control program for a tendon-driven soft exosuit, including sensor reception, motor commands and feedback handling.
- Built a tendon-driven soft exosuit pipeline with 10-ms end-to-end processing from sensor measurements to motor commands using multithreaded CAN communication.
- Assembled sensor board modules and connected the load cells, IMUs and motor to the control pipeline; observed motor-response delay during concurrent sensor communication.

Graduate Researcher

Korea University, Human-Machine Systems Lab | Seoul, South Korea | Advisor: Prof. Shinsuk Park | Sep. 2022 - Dec. 2024

Monocular Vision-Based Assistance Interface for Indoor Drones

- Developed a monocular vision-based pilot assistance interface to reduce human error during indoor UAV operation.
- Applied monocular depth estimation for obstacle perception using a single RGB camera on a lightweight UAV.
- Evaluated the interface with six participants in simulation: NASA-TLX mental demand decreased by approximately 43%, and mean System Usability Scale scores increased from 62.5 to 80 versus the RGB interface.

Visiting Researcher

Drexel University, Intelligent Machine Perception and Learning (IMAPLE) Lab | Philadelphia, PA, USA | Advisor: Prof. David Han | Sep. 2023 - Feb. 2024

- Applied CycleGAN-based domain transfer to synthetic indoor RGB images for monocular depth model training.
- Compared original and domain-transferred synthetic training data across three depth models; observed modest AbsRel improvements with mixed changes in other metrics. Translation training included NYU test RGB images.
- Developed a helium-supported aerial robot with Raspberry Pi-ESP32 hardware and laptop-based YOLOv5 tracking for video streaming and motor commands.

Industry Experience

Software Engineer, Robot Software Development

Samsung Electronics, Robot Vacuum Cleaner Development Lab | Suwon, South Korea | May 2026 - Present

Hardware-in-the-Loop Simulation and Validation

- Built an Isaac Sim-based HIL test system with Samsung R&D Institute-Delhi, integrating a physical controller board and production control software with simulated robot sensors.

Mechanical Engineer, Robot Product Development

Samsung Electronics, Robot Vacuum Cleaner Development Lab | Suwon, South Korea | Feb. 2025 - Apr. 2026

Robot Vacuum Cleaner Mechanism Development and Mass Production

- Designed and validated mop actuation mechanisms for robot vacuum cleaners, including durability improvement.
- Supported mass production and quality validation, including performance testing and vision-based inspection.

Publications & Presentations

DART: From Attentive Reasoning to Intuitive Execution for Experience-Driven Robot Intelligence

Seungyeop Lee^{*}, Yeeun Kim^{*}, Sooho Park, Seong Oh Lee, Jong Jin Park[†]

Samsung Electronics Co., Ltd., Suwon, Republic of Korea

Submitted to the 2027 IEEE International Conference on Robotics and Automation (ICRA) | Under review

[PDF](#)

Domain-Transferred Synthetic Data Generation for Improving Monocular Depth Estimation

Knut Peterson^{*†}, Seungyeop Lee^{*}, Solmaz Arezoomandan, David Han

Drexel University · Korea University

2025 25th International Conference on Control, Automation and Systems (ICCAS) | 2025

[PDF](#)

A Development of Intuitive Controller and Monocular Vision based Interface for Disaster Response UAV

Seungyeop Lee

Department of Mechanical Engineering, Korea University

Master's thesis, Department of Mechanical Engineering, Korea University | Advisor: Prof. Shinsuk Park | Feb. 2025

[PDF](#)

Micro UAV Autonomous Navigation System With Deep Learning based Monocular Vision Depth Estimation

Seungyeop Lee, Shinsuk Park

Department of Mechanical Engineering, Korea University

Institute of Control, Robotics and Systems Conference (ICROS) | 2024 | Poster presentation.

[PDF](#) | [Poster](#)

Development of ROS based indoor autonomous flight Quadcopter Platform using Visual Odometry

Seungyeop Lee, Shinsuk Park

Department of Mechanical Engineering, Korea University

Institute of Control, Robotics and Systems Conference (ICROS) | 2023 | Poster presentation.

[PDF](#)

Selected Projects & Leadership

Horang-Nabi Model Aircraft Club Foundation

Founder & President | Korea University | South Korea | Aug. 2019 - Dec. 2021

- Founded Korea University's first model aircraft club and led UAV projects, operations, and undergraduate study activities.

Korea Robot Aircraft Competition

Computer Vision Development | Korea Aerospace Industries Association | South Korea | Apr. 2021 - Sep. 2021

- Developed a pedestrian-tracking UAV for a safe-walk companion service, combining autonomous user tracking with onboard perception and flight control.

National University Student Model Aircraft Competition

Team Leader | Gyeongsang National University | South Korea | Nov. 2019 - Nov. 2020

- Led a team to design and build a scale aircraft for the 21st national university model aircraft competition.

Social Economy Engineering Ideas Competition

Mechanical Design | University of Seoul | South Korea | Sep. 2020 - Oct. 2020

- Proposed a foldable stair-climbing mobility aid to improve subway accessibility for elderly and mobility-limited users.

Pi Village Start-up Team - DRONEDU

Founder | Korea University | South Korea | Jan. 2020 - Oct. 2020

- Developed a drone education program for high school and university students as a campus start-up initiative.

Awards and Honors

Intelligent Robotics Overseas Research Fellowship

Sep. 2023 - Feb. 2024

Funded by the South Korean government through the Ministry of Trade, Industry and Energy (MOTIE).

Excellence Award, Future Automobile Industry-University Assignment Project Competition

Inter-university Alliance for Future Vehicular Technology (I4FT), South Korea | Jan. 2023

Third Prize, National University Student Model Aircraft Competition

Gyeongsang National University, South Korea | Nov. 2020

Third Prize, Social Economy Engineering Idea Competition

University of Seoul, South Korea | Oct. 2020

Third Prize, Creative Comprehensive Design Competition

Innovation Center for Engineering Education, Korea University, South Korea | Sep. 2020

First Prize, 22nd Pi Village Start-up Team - DRONEDU

Korea University, South Korea | Jun. 2020

Skills and Techniques

Hardware

- 3D CAD (NX, Fusion 360)
- Rapid prototyping
- Laser cutting
- 3D printing

Programming

- Python
- C++
- ROS
- MATLAB
- Microprocessors (Jetson, Arduino, Raspberry Pi)

Teaching and Mentoring

Teaching Assistant, Dynamics

Fall 2024

- Taught two-link manipulator modeling using Adams simulation.

Teaching Assistant, Robotics

Spring 2023, Spring 2024

- Taught Adams simulation, Raspberry Pi, and PID-based DC motor control.

Robotics Career Mentor, SC Joint Career Consulting

Mar. 17, 2026

- Participated in a government-supported career mentoring event as a Samsung Electronics engineer.

References

Dr. Shinsuk Park

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Dr. David Han

Professor, Department of Electrical and Computer Engineering | Drexel University

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Dr. Jinsoo Kim

Assistant Professor, Department of Electrical and Computer Engineering | Seoul National University

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